

SELECT within SELECT Tutorial

Language: English • 日本語 • 中文

This tutorial looks at how we can use SELECT statements within SELECT statements to perform more complex queries.

name	continent	area	population	gdp
Afghanistan	Asia	652230	25500100	20343000000
Albania	Europe	28748	2831741	12960000000
Algeria	Africa	2381741	37100000	188681000000
Andorra	Europe	468	78115	3712000000
Angola	Africa	1246700	20609294	100990000000
...				

Using nested SELECT

Summary

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1.

Bigger than Russia

List each country name where the population is larger than that of 'Russia'.

```
world(name, continent, area, population, gdp)
```

```
SELECT name
FROM world
WHERE population >
      (SELECT population
       FROM world
       WHERE name = 'Russia');
```

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Correct answer

name
Bangladesh
Brazil
China
India
Indonesia
Nigeria
Pakistan
United States

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2.

Richer than UK

Show the countries in Europe with a per capita GDP greater than 'United Kingdom'.

Per Capita GDP

```
SELECT name FROM world
WHERE continent = 'Europe'
  AND gdp/population >
      (SELECT gdp/population
       FROM world
       WHERE name = 'United Kingdom');
```

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Correct answer

name
Andorra
Austria
Belgium

Denmark
Finland
France
Germany

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3.

Neighbours of Argentina and Australia

List the name and continent of countries in the continents containing either Argentina or Australia. Order by name of the country.

```
SELECT name, continent FROM world

WHERE continent IN
  (SELECT continent
   FROM world
   WHERE name IN ('Argentina', 'Australia'))
ORDER BY name;
```

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Correct answer

name	continent
Argentina	South America
Australia	Oceania
Bolivia	South America
Brazil	South America
Chile	South America
Colombia	South America
Ecuador	South America
Peru	South America

4.

Between Canada and Poland

Which country has a population that is more than United Kingdom but less than Germany? Show the name and the population.

```
SELECT name, population FROM world

WHERE population >
  (SELECT population
   FROM world
   WHERE name = 'United Kingdom')
AND population <
  (SELECT population
```

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Correct answer

name	population
Congo, Democratic Republic of	69360000
France	65906000
Iran	77552000
Thailand	64456700
Turkey	76667864

5.

Percentages of Germany

Germany (population roughly 80 million) has the largest population of the countries in Europe. Austria (population 8.5 million) has 11% of the population of Germany.

Show the name and the population of each country in Europe. Show the population as a percentage of the population of Germany.

The format should be *Name, Percentage* for example:

name	percentage
Albania	3%
Andorra	0%
Austria	11%

...	...
-----	-----

Decimal places

Percent symbol %

```
SELECT name,  
       CONCAT(ROUND(population * 100 /  
                    (SELECT population  
                      FROM world  
                      WHERE name = 'Germany')), '%') AS percentage  
FROM world
```

Submit SQL

restore default

Correct answer

name	percentage
Albania	3%
Andorra	0%
Austria	11%
Belarus	12%
Belgium	14%
Bosnia and Herzegovina	5%
Bulgaria	9%
Germany	50%

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To get a well rounded view of the important features of SQL you should move on to the next tutorial concerning aggregates.

To gain an absurdly detailed view of one insignificant feature of the language, read on.

We can use the word **ALL** to allow **>=** or **>** or **<** or **<=** to act over a list. For example, you can find the largest country in the world, by population with this query:

```
SELECT name
FROM world
WHERE population >= ALL(SELECT population
                        FROM world
                        WHERE population>0)
```

You need the condition **population>0** in the sub-query as some countries have **null** for population.

6.

Bigger than every country in Europe

Which countries have a GDP greater than every country in Europe? [Give the name only.] (Some countries may have NULL gdp values)


```
SELECT name FROM world
WHERE gdp > (SELECT MAX(gdp)
             FROM world
             WHERE continent = 'Europe');
```

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Correct answer

name
China
Japan
United States

We can refer to values in the outer SELECT within the inner SELECT. We can name the tables so that we can tell the difference between the inner and outer versions.

7.

Largest in each continent

Find the largest country (by area) in each continent, show the continent, the name and the area:

```
SELECT continent, name, area FROM world x

WHERE area >= ALL(
    SELECT area
    FROM world y
    WHERE x.continent = y.continent
    AND y.area > 0
);
```

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Correct answer

continent	name	area
Africa	Algeria	2381741
Oceania	Australia	7692024
South America	Brazil	8515767
North America	Canada	9984670
Asia	China	9596961
Caribbean	Cuba	109884
Europe	Kazakhstan	2724900
Europe	Russia	17125312

The above example is known as a **correlated** or **synchronized** sub-query.

Using correlated subqueries

8.

First country of each continent (alphabetically)

List each continent and the name of the country that comes first alphabetically.

```
SELECT continent, name FROM world x
WHERE name <= ALL(
  SELECT name
  FROM world y
  WHERE x.continent = y.continent
);
```

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Correct answer

continent	name
Africa	Algeria
Asia	Afghanistan
Caribbean	Antigua and Barbuda
Eurasia	Armenia
Europe	Albania
North America	Belize
Oceania	Australia
South America	Argentina

9.

Difficult Questions That Utilize Techniques Not Covered In Prior Sections

Find the continents where all countries have a population <= 25000000. Then find the names of the countries associated with these continents. Show name, continent and population.

```
SELECT name, continent, population
FROM world
WHERE continent IN (
    SELECT continent
    FROM world
    GROUP BY continent
    HAVING MAX(population) <= 25000000
);
```

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Correct answer

name	continent	population
Antigua and Barbuda	Caribbean	86295
Australia	Oceania	23545500
Bahamas	Caribbean	351461
Barbados	Caribbean	285000
Cuba	Caribbean	11167325
Dominica	Caribbean	71293
Dominican Republic	Caribbean	9445281
...	...	...

1.

Three time bigger

Some countries have populations more than three times that of all of their neighbours (in the same continent). Give the countries and continents.

```
SELECT w1.name, w1.continent
FROM world w1
WHERE w1.population > ALL (
  SELECT 3 * w2.population
  FROM world w2
  WHERE w1.continent = w2.continent
  AND w1.name <> w2.name
);
```

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Correct answer

name	continent
Australia	Oceania
Brazil	South America
Russia	Eurasia

Nested SELECT Quiz

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